



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-19

Experiment 19. Prolog Program for Planets Database

Aim

To write a Prolog program to create a **database of planets** and retrieve their information.

Objectives

- To store planet information using facts.
- To understand Prolog databases.
- To retrieve planet details using queries.

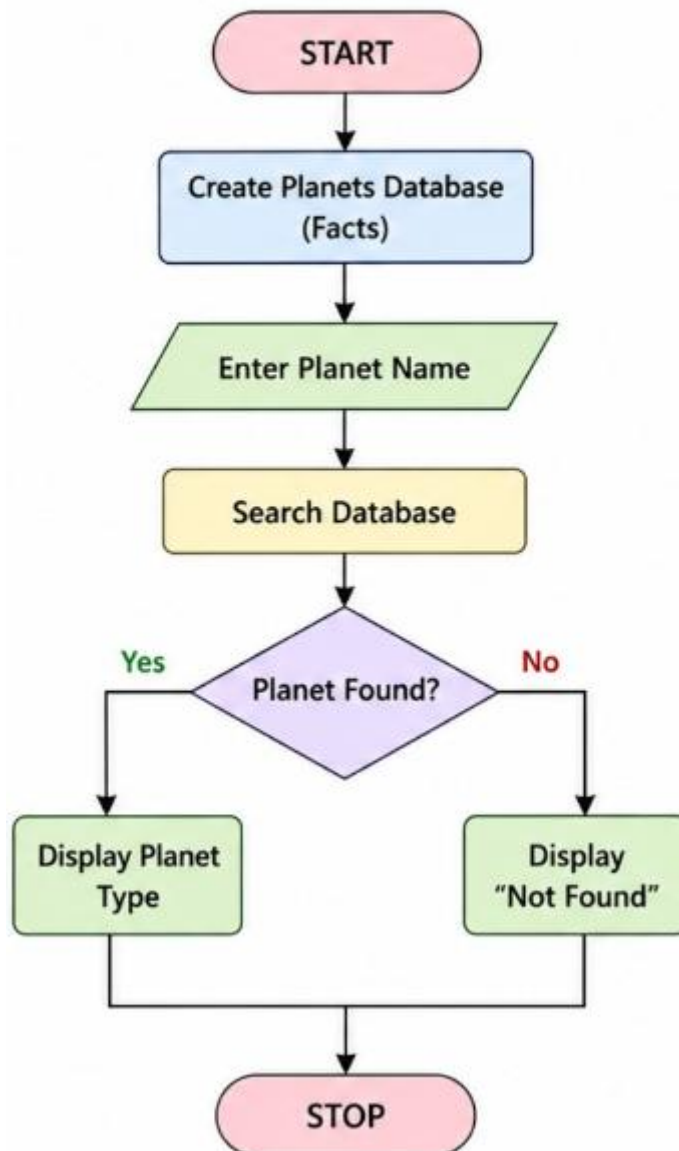
Algorithm

1. Create facts for different planets.
2. Store their types.
3. Enter a planet name.
4. Search the database.
5. Display the planet type.

Flowchart

19. PROLOG PROGRAM FOR PLANETS DATABASE

FLOWCHART



Code

```
planet(mercury, terrestrial).
```

```
planet(venus, terrestrial).
```

```
planet(earth, terrestrial).
```

```
planet(mars, terrestrial).
```

```
planet(jupiter, gas_giant).
```

```
planet(saturn, gas_giant).
```

```
planet(uranus, ice_giant).
```

```
planet(neptune, ice_giant).
```

```
find_planet(Name, Type) :-
```

```
    planet(Name, Type).
```

```
main :-
```

```
    Name = earth,
```

```
    find_planet(Name,
```

```
    Type),
```

```
    write('Planet: '),
```

```
    write(Name), nl,
```

```
    write('Type: '),  
    write(Type), nl.  
  
:- initialization(main).
```

Output



```
1 planet(mercury, terrestrial).  
2 planet(venus, terrestrial).  
3 planet(earth, terrestrial).  
4 planet(mars, terrestrial).  
5 planet(jupiter, gas_giant).  
6 planet(saturn, gas_giant).  
7 planet(uranus, ice_giant).  
8 planet(neptune, ice_giant).  
9  
10 find_planet(Name, Type) :-  
11     planet(Name, Type).  
12  
13  
14 main :-  
15     Name = earth,  
16     find_planet(Name, Type),  
17     write('Planet: '), write(Name), nl,  
18     write('Type: '), write(Type), nl.  
19  
20 :- initialization(main).
```

Output:
Planet: earth
Type: terrestrial

Result

Type = terrestrial.

Conclusion

Thus, the Prolog program for the planets database was successfully implemented.